

APP Layer — Part B Test Sequence

ALL STEPS PASSED

Modules: UILOC (LCD + keypad) · UIREM (UART terminal) · full super-loop, all 8 APP modules

Firmware test_app (branch phase2-app) · Proteus run 2026-10-04 · Tester: Ahmad

13 test blocks	82 checked steps	8 APP modules live	0 failures
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This sequence verifies the complete APP layer through its two user interfaces — the LCD/keypad (UILOC) and the UART terminal (UIREM) — running in the full super-loop. The hardware paths (lamps, AC, heater, door) were verified in part A; here the focus is accounts, permissions, arbitration, events, persistence, lockdown and storage. Blocks build on each other; the RESET-heavy blocks come last. Step IDs in brackets refer to **docs/test_plan.md** §3.

0 Setup

Part	Setting
ATmega32	Program File = <code>.pio/build/test_app/firmware.hex</code> · Clock Frequency = 16 MHz
Virtual Terminal	9600 baud, 8N1 · “Echo Typed Characters” off (the MCU echoes)
Instruments	I ² C debugger on SDA/SCL · scope A = PB3 (dimmer), B = PD4 (fan), C = PD5 (servo)
LM35 start values	Ambient (PA0) = 25 °C · Water (PA1) = 40 °C
Keypad keys	= Enter · C or * back / delete · + - up / down

Login counters. Wrong logins add up **per source** (terminal and keypad separately) until a correct login on that source; the 3rd locks the system. Outside block 10, follow any deliberate failure with one correct login.

Text styles. `monospace blue` = terminal input / output · `monospace brown` = LCD text and keypad keys.

1 Automatic part and boot

[T-33]

#	Do	Expect	Result
A-1	Play, touch nothing	All [PASS], SUMMARY: ... 0 failed, then LIVE: the full system runs now ...	PASS
A-2	Read the terminal	==== SMART HOME + WATER HEATER ==== then Hey, please enter your username:	PASS
A-3	Look at the LCD	Pages alternate every 3 s: Lamps:----- D0% / AC:OFF Room:25C, then Heater:OFF / Water:40C Set:..	PASS
A-4	Type <code>abc</code> , Backspace, <code>d</code>	Echo shows <code>abd</code> ; Backspace removes one character	PASS

2 Known starting state: factory reset

[T-01, T-21, T-76] The EEPROM still holds accounts from earlier tests.

#	Do	Expect	Result
F-1	Enter, then admin / 1234	Password echoes ****; Welcome admin. You are logged in as ADMIN.; admin menu 1-16; prompt admin>	PASS
F-2	16, then YES	[OK] Factory reset done. Admin is admin / 1234., then the banner (logged out)	PASS
F-3	I ² C debugger	13 page writes, from 0xC0 down to 0x00	PASS
F-4	Pause, open the 24C08 memory window	0x00 = A5 · 0x01 = 01 · 0x10 = 3C · 0x20 = admin · 0x28 = 1234 · 0x30-0xCF = 00	PASS
F-5	Resume, log in admin / 1234	Works	PASS

3 Accounts

[T-02, T-03, T-05, T-06]

#	Do	Expect	Result
U-1	10 → bob → 5678	Prompts New remote username (1-8 characters): / Password (4-8 characters): (echo *); [OK] Remote user bob added.	PASS
U-2	12 → 1111 → 2222	[OK] Keypad user 1111 added.	PASS
U-3	14	Remote users (1 of 5): bob · Keypad users (1 of 5): 1111 · no passwords	PASS
U-4	10 with bob again	[ERR] This username already exists. + menu	PASS
U-5	10 with u2, u3, u4, u5 (pw pass1), then a 6th name	Four [OK], then [ERR] The user list is full (5).	PASS
U-6	12 with ID 12ab	[ERR] Keypad users need digits only.	PASS
U-7	11 u5	[OK] Remote user u5 removed. — frees a slot so U-8 reaches the password check	PASS
U-8	10 with name x, password 12	[ERR] Password must be 4 to 8 characters.	PASS
U-9	11 nobody	[ERR] User not found.	PASS
U-10	Keypad: ID 1234, PIN 1234	Wrong ID or PIN / 2 tries left — the admin cannot log in on the keypad	PASS
U-11	Keypad: 1111 = 2222 =, then C	Menu, then status screen; keypad counter back to 3	PASS

4 Terminal commands and errors

[T-36 – T-39, T-41 – T-46, T-70] Logged in as admin.

#	Do	Expect	Result
R-1	1 3	[OK] Lamp 3 is now ON; lamp 3 lights; only the prompt follows	PASS
R-2	1, then 3 at the argument prompt	[OK] Lamp 3 is now OFF	PASS
R-3	2 70	[OK] Dimmer is now 70%; scope A $\approx$ 70 %	PASS
R-4	3	AC is OFF. Room temperature 25C.	PASS
R-5	4	Heater is OFF. Water 40C, set 60C.	PASS
R-6	5 65	[OK] Heater set temperature is now 65C; one page write on the I ² C debugger	PASS
R-7	7, 7, 8, 8	[OK] Door is now OPEN, Door is already open., [OK] Door is now CLOSED, Door is already closed.	PASS
R-8	6	Status block incl. Dimmer: 70%, set 65C, Door : CLOSED, Keypad: BLOCKED, nobody logged in	PASS
R-9	99, then abc	[ERR] Unknown command. twice, each with the menu	PASS
R-10	A 20-character line	[ERR] Input too long. + menu	PASS
R-11	1 9, 2 55, 5 62	Lamp, dimmer and temperature range errors, each with the menu	PASS
R-12	?, then an empty Enter	The menu both times	PASS
R-13	0	Goodbye admin., then the banner	PASS

5 Remote user permissions

[T-13, T-15, T-36, T-39]

#	Do	Expect	Result
P-1	Log in bob / 5678	Welcome bob. You are logged in as USER.; menu 1-6 only; prompt bob>	PASS
P-2	7, 8, 9, 10, 14, 16	Each: [ERR] Not allowed for your role.; servo does not move	PASS
P-3	5 50	[OK] Heater set temperature is now 50C (allowed, D-12)	PASS
P-4	6	Status block without the Keypad line	PASS
P-5	Log in again with a password typo + Backspace	One * removed; corrected password accepted	PASS

6 Keypad and LCD

[T-22 – T-32] Log out of the terminal first.

#	Do	Expect	Result
K-1	Press 1	User ID:1 — the first digit is kept	PASS
K-2	11, *, 1	* deletes one digit; ID shown in clear 1111	PASS
K-3	=, then 2222	PIN:**** — every digit as *	PASS
K-4	=	1Lamps 2Dimmer / 3AC 4Heat C=Exit — no door entry	PASS
K-5	1, then 1...5, then C	Lamp 1 2 3 4 5 marks * / - follow the LEDs; C back to the menu	PASS
K-6	2, + x3, - x1, C	Dimmer: 10, 20, 30, 20 %; lamp follows	PASS
K-7	3; ambient LM35 → 27	AC:OFF Room:27C / on>28 off<21; updates within ≈ 1 s	PASS
K-8	4; +, then -	Water:40C OFF / Set:50C + / -; 55, then 50	PASS
K-9	C, C	Menu, then logged out to the status screen	PASS
K-10	At an empty User ID: press C	Back to the status screen	PASS
K-11	Log in, press nothing for 30 s	Automatic logout	PASS
K-12	Lamps 1+3, dimmer 70, ambient 30, heater ON; not logged in	Lamps:1-3-- D70% / AC:ON Room:30C, then Heater:HEATING / Water:...C Set:50	PASS

7 Terminal and keypad arbitration

[T-16 – T-20]

#	Do	Expect	Result
X-1	Terminal: bob. Keypad: 1111, toggle lamp 2	Keypad login works; terminal: [INFO] Keypad user logged in, [INFO] Lamp 2 is now ON	PASS
X-2	Terminal: 0, log in as admin	Keypad session ends at once (status screen)	PASS
X-3	Press any keypad key	Keypad blocked / by admin for 2 s, then status	PASS
X-4	Admin: 9	[OK] Keypad control is now ALLOWED; keypad login works	PASS
X-5	Admin: 9 again	... BLOCKED; LCD drops back to the status screen	PASS
X-6	Admin: send nothing for 120 s	[INFO] No input for 120 s. Logged out. + banner; keypad works again	PASS

8 Events

[T-34, T-35, T-72] Logged in as admin, keypad allowed (9).

#	Do	Expect	Result
V-1	Keypad user toggles lamp 4	[INFO] Lamp 4 is now ON, then the prompt again	PASS
V-2	Heater ON/OFF button	[INFO] Heater is now ON, then [INFO] Heater: HEATING or IDLE	PASS
V-3	Panel: Up, Up, wait 5 s	[INFO] Heater set temperature is now 60C	PASS
V-4	Ambient → 29	[INFO] AC is now ON	PASS
V-5	Type 1 (no Enter), press heater ON/OFF, then Enter, 3	No [INFO] while the line is half-typed; it appears after the command	PASS
V-6	1 2 from the terminal	Only [OK] ..., no duplicate [INFO]	PASS

9 Persistence

[T-07, T-61]

#	Do	Expect	Result
M-1	Press RESET	Automatic part 0 failed; banner; LCD status pages	PASS
M-2	Terminal: bob / 5678	Works	PASS
M-3	Keypad: 1111 / 2222	Works	PASS
M-4	Terminal: 4	set 60C — the last set value survived	PASS

10 Login counters and lockdown

[T-08 – T-12, T-73, T-74]

#	Do	Expect	Result
L-1	Terminal: 2 wrong, 1 correct (bob), 0, 2 wrong	2 attempts left, 1 attempt left, login, then 2 / 1 again — no lockdown	PASS
L-2	RESET. Terminal 2 wrong; keypad 2 wrong	No lockdown — separate counters	PASS
L-3	RESET. Heater heating, ambient 29, lamps 1+2, dimmer 50, door open, 0	Setup for L-4	PASS
L-4	Terminal: 3 wrong pairs	2 attempts left., 1 attempt left., then [ALERT] 3 wrong logins. SYSTEM LOCKED. Reset to restart.	PASS
L-5	Check outputs	Buzzer 0.5 / 0.5 s · red/blue off · fan off · 7-seg blank · LCD SYSTEM LOCKED / Reset required · lamps, dimmer, door unchanged	PASS
L-6	Type admin/1234; keypad keys; heater buttons	Nothing answered, nothing happens	PASS
L-7	RESET. Keypad: 3 wrong ID/PIN pairs	2 tries left, 1 tries left, SYSTEM LOCKED; buzzer; terminal prints [ALERT] with nobody logged in	PASS
L-8	RESET	Normal start, buzzer silent, login works	PASS

11 Storage

[T-77, T-78]

#	Do	Expect	Result
S-1	Admin: heater blinking in setting mode; add 3 users quickly (10)	One 16-byte page write per user; blinking and keypad never stall	PASS
S-2	Stop. 24C08 → Exclude from Simulation . Play	Starts normally, no hang	PASS
S-3	Log in admin / 1234	Works from RAM; [WARN] EEPROM not responding. Changes will not be saved.	PASS
S-4	Stop, include the 24C08 again	—	PASS

12 Admin password and final reset

[T-21]

#	Do	Expect	Result
Z-1	Admin: 15 → abcd	Echo ***, [OK] Admin password changed.	PASS
Z-2	0; log in with 1234, then abcd	1234 refused (2 attempts left); abcd accepted	PASS
Z-3	16, then no	Cancelled.	PASS
Z-4	16, then YES	Factory reset; admin / 1234 works; 14 shows empty lists	PASS

Result — 2026-10-04. All blocks 0-12 executed in Proteus on the full schematic; all 82 steps matched their expectation. Together with part A (48 steps), the whole APP layer — LIGHT, DOOR, CLIMATE, HEATER, ALARM, SEC, UILOC, UIREM — is verified. Next: Phase 3 (src/APP/main.c and the full test_plan.md run).